Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.20)

Metric

$$ds^2 = \left(\frac{\gamma-6}{4\,M\gamma^2}\right)\mathrm{d}t\otimes\mathrm{d}t + \left(-\frac{(\gamma-6)t^{\frac{2}{\gamma}-1}}{4\,M\gamma^2}\right)\mathrm{d}x\otimes\mathrm{d}x + \left(-\frac{\sqrt{\gamma-2}(\gamma-6)\sqrt{-3\,\gamma+2}t^{\frac{2}{\gamma}-1}z}{8\,M\gamma^3}\right)\mathrm{d}x\otimes\mathrm{d}y + \left(-\frac{\sqrt{\gamma-2}(\gamma-6)\sqrt{-3\,\gamma+2}t^{\frac{2}{\gamma}-1}z}{8\,M\gamma^3}\right)\mathrm{d}y \otimes\mathrm{d}x + \left(\frac{\left(3\,\gamma^3-26\,\gamma^2+52\,\gamma-24\right)t^{\frac{2}{\gamma}}z^2-4\left(\gamma^3-6\,\gamma^2\right)t^{\frac{1}{\gamma}+\frac{3}{2}}}{16\,M\gamma^4t}\right)\mathrm{d}y\otimes\mathrm{d}y + \left(-\frac{(\gamma-6)t^{\frac{1}{\gamma}+\frac{1}{2}}}{4\,M\gamma^2}\right)\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}}\right)dt + \left(\frac{\sqrt{2}t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}}\right)dz$$

$$n = \left(\frac{\sqrt{2}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}}\right)dt + \left(-\frac{\sqrt{2}t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}}\right)dz$$

$$m = \left(\frac{\sqrt{2}t^{\left(\frac{1}{\gamma}\right)}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}\sqrt{t}}\right)dx + \left(\frac{i\sqrt{2}t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}} + \frac{\sqrt{2}\sqrt{-3\,\gamma^2+8\,\gamma-4}t^{\left(\frac{1}{\gamma}\right)}z}{8\sqrt{M}\gamma\sqrt{-\frac{\gamma^2}{\gamma-6}}\sqrt{t}}\right)dy$$

$$\overline{m} = \left(\frac{\sqrt{2}t^{\left(\frac{1}{\gamma}\right)}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}\sqrt{t}}\right)dx + \left(-\frac{i\sqrt{2}t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}{4\sqrt{M}\sqrt{-\frac{\gamma^2}{\gamma-6}}} + \frac{\sqrt{2}\sqrt{-3\,\gamma^2+8\,\gamma-4}t^{\left(\frac{1}{\gamma}\right)}z}{8\sqrt{M}\gamma\sqrt{-\frac{\gamma^2}{\gamma-6}}\sqrt{t}}\right)dy$$

Contravariant components

$$l = \left(\frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{\gamma-6}\right) dt + \left(-\frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\right) dz$$

$$n = \left(\frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{\gamma-6}\right) dt + \left(\frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\right) dz$$

$$m = \left(\frac{(3i+3)\ \sqrt{2}\sqrt{M}\gamma^2 z^2\sqrt{|\gamma-6|}}{4\,(\gamma^2-6\,\gamma)t} - \frac{\sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{2\,(\gamma^2-6\,\gamma)t^{\frac{1}{4}}t^{\left(\frac{1}{\gamma}\right)}} - \frac{(2i+2)\ \sqrt{2}\sqrt{M}\gamma z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t} - \frac{\sqrt{2}\sqrt{M}\gamma^2\sqrt{t}\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t^{\left(\frac{1}{\gamma}\right)}} + \frac{(i+1)\ \sqrt{2}\sqrt{M}z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}\right) dx$$
$$+ \left(\frac{(i-1)\ \sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}z\sqrt{|\gamma-6|}}{2\,(\gamma-6)t} - \frac{i\,\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\right) dy$$

$$\overline{m} = \left(\frac{(3i+3)\ \sqrt{2}\sqrt{M}\gamma^2 z^2\sqrt{|\gamma-6|}}{4\,(\gamma^2-6\,\gamma)t} + \frac{\sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{2\,(\gamma^2-6\,\gamma)t^{\frac{1}{4}}t^{\left(\frac{1}{\gamma}\right)}} - \frac{(2i+2)\ \sqrt{2}\sqrt{M}\gamma z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t} - \frac{\sqrt{2}\sqrt{M}\gamma^2\sqrt{t}\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t^{\left(\frac{1}{\gamma}\right)}} + \frac{(i+1)\ \sqrt{2}\sqrt{M}z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}\right) dx$$
$$+ \left(\frac{(i-1)\ \sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}z\sqrt{|\gamma-6|}}{2\,(\gamma-6)t} + \frac{i\,\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\right) dy$$

Directional Derivatives

$$D = \frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{\gamma-6}\,\partial_t - \frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\,\partial_z$$

$$\Delta = \frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{\gamma-6}\,\partial_t + \frac{\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\,\partial_z$$

$$\delta = \frac{(3i+3)\ \sqrt{2}\sqrt{M}\gamma^2 z^2\sqrt{|\gamma-6|}}{4\,(\gamma^2-6\,\gamma)t} - \frac{\sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{2\,(\gamma^2-6\,\gamma)t^{\frac{1}{4}}t^{\left(\frac{1}{\gamma}\right)}} - \frac{(2i+2)\ \sqrt{2}\sqrt{M}\gamma z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}$$
$$- \frac{\sqrt{2}\sqrt{M}\gamma^2\sqrt{t}\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t^{\left(\frac{1}{\gamma}\right)}} + \frac{(i+1)\ \sqrt{2}\sqrt{M}z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}\,\partial_x + \frac{(i-1)\ \sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}z\sqrt{|\gamma-6|}}{2\,(\gamma-6)t} - \frac{i\,\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}}t^{\frac{1}{2\gamma}}}\,\partial_y$$

$$\overline{\delta} = \frac{(3i+3)\ \sqrt{2}\sqrt{M}\gamma^2 z^2\sqrt{|\gamma-6|}}{4\,(\gamma^2-6\,\gamma)t} + \frac{\sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z\sqrt{|\gamma-6|}}{2\,(\gamma^2-6\,\gamma)t^{\frac{1}{4}} t^{\left(\frac{1}{\gamma}\right)}} - \frac{(2i+2)\ \sqrt{2}\sqrt{M}\gamma z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}$$
$$- \frac{\sqrt{2}\sqrt{M}\gamma^2\sqrt{t}\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t^{\left(\frac{1}{\gamma}\right)}} + \frac{(i+1)\ \sqrt{2}\sqrt{M}z^2\sqrt{|\gamma-6|}}{(\gamma^2-6\,\gamma)t}\,\partial_x + \frac{(i-1)\ \sqrt{2}\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}z\sqrt{|\gamma-6|}}{2\,(\gamma-6)t} + \frac{i\,\sqrt{2}\sqrt{M}\gamma\sqrt{|\gamma-6|}}{(\gamma-6)t^{\frac{1}{4}} t^{\frac{1}{2\,\gamma}}}\,\partial_y$$

Spin Coefficients

$$\rho = -\frac{3i\,\sqrt{M}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{8\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} - \frac{(3i+3)\ \sqrt{M}\gamma^3 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{i\,\sqrt{M}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$+ \frac{(2i+2)\ \sqrt{M}\gamma^2 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{\sqrt{M}\gamma^3|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} + \frac{3i\,\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$- \frac{(i+1)\ \sqrt{M}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{3\,\sqrt{M}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} - \frac{i\,\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$

$$\sigma = -\frac{3i\,\sqrt{M}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{8\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} + \frac{(i+1)\ \sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{(3i+3)\ \sqrt{M}\gamma^3 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{i\,\sqrt{M}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$+ \frac{(i+1)\ \sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{(2i+2)\ \sqrt{M}\gamma^2 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{\sqrt{M}\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} - \frac{3\,\sqrt{M}\gamma^3|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t}$$
$$+ \frac{3i\,\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} - \frac{(i+1)\ \sqrt{M}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{\sqrt{M}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} - \frac{i\,\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$

$$\kappa = 0$$

$$\mu = \frac{3i\,\sqrt{M}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{8\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} - \frac{(3i+3)\ \sqrt{M}\gamma^3 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{i\,\sqrt{M}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$+ \frac{(2i+2)\ \sqrt{M}\gamma^2 t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{\sqrt{M}\gamma^3|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} - \frac{3i\,\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$- \frac{(i+1)\ \sqrt{M}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{3\,\sqrt{M}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t} + \frac{i\,\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4-12\,\sqrt{2}\gamma^3+36\,\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$

$$\lambda = \frac{3i\sqrt{M}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2 |\gamma-6|^{\frac{3}{2}}}{8\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} + \frac{(i+1)\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^2 t^{\frac{1}{2\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{(3i+3)\sqrt{M}\gamma^3 t^{\frac{1}{2\gamma}} z|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{i\sqrt{M}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2 |\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$
$$+ \frac{(i+1)\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\gamma}} z|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} + \frac{(2i+2)\sqrt{M}\gamma^2 t^{\frac{1}{2\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t} + \frac{3\sqrt{M}\gamma^3|\gamma-6|^{\frac{3}{2}}}{4\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t}$$
$$- \frac{3i\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}} - \frac{(i+1)\sqrt{M}\gamma t^{\frac{1}{2\gamma}} z|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{7}{4}}} - \frac{\sqrt{M}\gamma^2|\gamma-6|^{\frac{3}{2}}}{2\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t} + \frac{i\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|^{\frac{3}{2}}}{\left(\sqrt{2}\gamma^4 - 12\sqrt{2}\gamma^3 + 36\sqrt{2}\gamma^2\right)t^{\frac{5}{2}}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{9i\sqrt{2}\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,t^{\frac{5}{2}}\sqrt{|\gamma-6|}} - \frac{(3i+3)\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}t^{\frac{1}{2\gamma}} z}{32\,t^{\frac{7}{4}}\sqrt{|\gamma-6|}} - \frac{3i\sqrt{2}\sqrt{M}\gamma t^{\frac{1}{2\gamma}} z}{8\,t^{\frac{7}{4}}\sqrt{|\gamma-6|}} + \frac{15i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,t^{\frac{5}{2}}\sqrt{|\gamma-6|}} + \frac{i\sqrt{2}\sqrt{M}t^{\frac{1}{2\gamma}} z}{t^{\frac{7}{4}}\sqrt{|\gamma-6|}}$$
$$+ \frac{(i+1)\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}t^{\frac{1}{2\gamma}} z}{16\,\gamma t^{\frac{7}{4}}\sqrt{|\gamma-6|}} - \frac{i\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}}{8\,t\sqrt{|\gamma-6|}} - \frac{\sqrt{2}\sqrt{M}\gamma}{8\,t\sqrt{|\gamma-6|}} - \frac{7i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,\gamma t^{\frac{5}{2}}\sqrt{|\gamma-6|}} - \frac{i\sqrt{2}\sqrt{M}t^{\frac{1}{2\gamma}} z}{2\,\gamma t^{\frac{7}{4}}\sqrt{|\gamma-6|}} - \frac{\sqrt{2}\sqrt{M}}{4\,t\sqrt{|\gamma-6|}} + \frac{i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,\gamma^2 t^{\frac{5}{2}}\sqrt{|\gamma-6|}}$$

$$\gamma = \frac{9i\sqrt{2}\sqrt{M}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,t^{\frac{5}{2}}\sqrt{|\gamma-6|}} - \frac{(3i+3)\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}t^{\frac{1}{2\gamma}} z}{32\,t^{\frac{7}{4}}\sqrt{|\gamma-6|}} - \frac{3i\sqrt{2}\sqrt{M}\gamma t^{\frac{1}{2\gamma}} z}{8\,t^{\frac{7}{4}}\sqrt{|\gamma-6|}} - \frac{15i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,t^{\frac{5}{2}}\sqrt{|\gamma-6|}} + \frac{i\sqrt{2}\sqrt{M}t^{\frac{1}{2\gamma}} z}{t^{\frac{7}{4}}\sqrt{|\gamma-6|}}$$
$$+ \frac{(i+1)\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}t^{\frac{1}{2\gamma}} z}{16\,\gamma t^{\frac{7}{4}}\sqrt{|\gamma-6|}} + \frac{i\sqrt{2}\sqrt{M}\sqrt{3\gamma-2}\sqrt{\gamma-2}}{8\,t\sqrt{|\gamma-6|}} + \frac{\sqrt{2}\sqrt{M}\gamma}{8\,t\sqrt{|\gamma-6|}} + \frac{7i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,\gamma t^{\frac{5}{2}}\sqrt{|\gamma-6|}} - \frac{i\sqrt{2}\sqrt{M}t^{\frac{1}{2\gamma}} z}{2\,\gamma t^{\frac{7}{4}}\sqrt{|\gamma-6|}} + \frac{\sqrt{2}\sqrt{M}}{4\,t\sqrt{|\gamma-6|}} - \frac{i\sqrt{2}\sqrt{M}t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,\gamma^2 t^{\frac{5}{2}}\sqrt{|\gamma-6|}}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{M\gamma^2}{4\,t^2|\gamma-6|} + \frac{3\,M\gamma}{2\,t^2|\gamma-6|}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{3i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}} - \frac{(i+1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{16\,(\gamma^2-6\,\gamma)t^{\frac{11}{4}}} - \frac{i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}} + \frac{i\,M t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{3i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\,t^{\frac{7}{2}}|\gamma-6|} - \frac{13i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,t^{\frac{7}{2}}|\gamma-6|} - \frac{M\gamma^2}{8\,t^2|\gamma-6|} + \frac{13i\,M t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,t^{\frac{7}{2}}|\gamma-6|} + \frac{3\,M\gamma}{4\,t^2|\gamma-6|} - \frac{3i\,M t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,\gamma t^{\frac{7}{2}}|\gamma-6|}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{3i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}} + \frac{(i+1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{16\,(\gamma^2-6\,\gamma)t^{\frac{11}{4}}} - \frac{i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}} + \frac{i\,M t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^2-6\,\gamma)t^{\frac{7}{2}}}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{M\gamma^2}{4\,t^2|\gamma-6|} + \frac{3\,M\gamma}{2\,t^2|\gamma-6|}$$

$$\Lambda = -\frac{M\gamma}{8\,t^2} + \frac{M}{6\,t^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{45i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{(15i+15)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} + \frac{(27i+27)\;M\gamma^4 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} - \frac{51i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} + \frac{(7i+7)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}}$$
$$- \frac{(45i+45)\;M\gamma^3 t^{\frac{1}{2\,\gamma}} z}{8\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} - \frac{9\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3}{8\,(\gamma^3-6\,\gamma^2)t^2} + \frac{15\,M\gamma^4}{8\,(\gamma^3-6\,\gamma^2)t^2} + \frac{10i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{(3i+3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}}$$
$$+ \frac{(21i+21)\;M\gamma^2 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} + \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2}{4\,(\gamma^3-6\,\gamma^2)t^2} - \frac{7\,M\gamma^3}{2\,(\gamma^3-6\,\gamma^2)t^2} - \frac{13i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{(3i+3)\;M\gamma t^{\frac{1}{2\,\gamma}} z}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} + \frac{3\,M\gamma^2}{2\,(\gamma^3-6\,\gamma^2)t^2} + \frac{3i\,M t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{15i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{17i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} + \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3}{8\,(\gamma^3-6\,\gamma^2)t^2} - \frac{5\,M\gamma^4}{8\,(\gamma^3-6\,\gamma^2)t^2} + \frac{10i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{3\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}}$$
$$- \frac{M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2}{4\,(\gamma^3-6\,\gamma^2)t^2} + \frac{7\,M\gamma^3}{6\,(\gamma^3-6\,\gamma^2)t^2} - \frac{13i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{6\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{M\gamma^2}{2\,(\gamma^3-6\,\gamma^2)t^2} + \frac{i\,M t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{45i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} + \frac{(15i+15)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} - \frac{(27i+27)\;M\gamma^4 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} - \frac{51i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} - \frac{(7i+7)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}}$$
$$+ \frac{(45i+45)\;M\gamma^3 t^{\frac{1}{2\,\gamma}} z}{8\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} - \frac{9\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3}{8\,(\gamma^3-6\,\gamma^2)t^2} + \frac{15\,M\gamma^4}{8\,(\gamma^3-6\,\gamma^2)t^2} + \frac{10i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} + \frac{(3i+3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}}$$
$$- \frac{(21i+21)\;M\gamma^2 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} + \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2}{4\,(\gamma^3-6\,\gamma^2)t^2} - \frac{7\,M\gamma^3}{2\,(\gamma^3-6\,\gamma^2)t^2} - \frac{13i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}} + \frac{(3i+3)\;M\gamma t^{\frac{1}{2\,\gamma}} z}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{11}{4}}} + \frac{3\,M\gamma^2}{2\,(\gamma^3-6\,\gamma^2)t^2} + \frac{3i\,M t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^3-6\,\gamma^2)t^{\frac{7}{2}}}$$

Newman-Penrose Equations

$$
\begin{aligned}
NP1 = & -\frac{45\,M\gamma^7 t^{\frac{2}{\gamma}} z^4}{128\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} + \frac{(63i-81)\;M\gamma^7 t^{\frac{3}{2\gamma}} z^3}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} + \frac{(135i-36)\;M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{15i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{(12i+9)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} \\
& - \frac{(33i+21)\;M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} + \frac{201\,M\gamma^6 t^{\frac{2}{\gamma}} z^4}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} - \frac{(165i-207)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} - \frac{(771i-204)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} \\
& - \frac{(38i+27)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} + \frac{(35i+31)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} - \frac{M\gamma^7}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{173\,M\gamma^5 t^{\frac{2}{\gamma}} z^4}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} - \frac{(3i+4)\;M\gamma^6|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} \\
& + \frac{(505i-591)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} + \frac{(703i-188)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{7i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{(39i+26)\;M\gamma^5 t^{\frac{1}{2\gamma}} z}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} + \frac{(i-11)\;M\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} \\
& + \frac{3\,M\gamma^6}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{143\,M\gamma^4 t^{\frac{2}{\gamma}} z^4}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} + \frac{(4i+7)\;M\gamma^5|\gamma-6|}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{(59i-49)\;M\gamma^4 t^{\frac{3}{2\gamma}} z^3}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} - \frac{(503i-148)\;M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \dots
\end{aligned}
$$

$$
\begin{aligned}
NP2 = & -\frac{45\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(3i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{(63i-81)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{15i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{15i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(129i-36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(i+1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\gamma}} z}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& - \frac{(3i-3)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(7i+19)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(9i+12)\;M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{33\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& - \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{(141i-171)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{131i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{7i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(173i-48)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(11i+11)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(i-4)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(5i+13)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(27i+16)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(2i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{(6i-3)\;M\gamma^7}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} \\
& + \frac{(3i-2)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{(6i-3)\;M\gamma^6|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{25\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP6 = & -\frac{81\,M\gamma^7 t^{\frac{2}{\gamma}} z^4}{128\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} - \frac{9i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{(39i+72)\;M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{105i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{513\,M\gamma^6 t^{\frac{2}{\gamma}} z^4}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} \\
& + \frac{21i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{(77i+204)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{101i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{1161\,M\gamma^5 t^{\frac{2}{\gamma}} z^4}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} \\
& + \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{3i\,M\gamma^6|\gamma-6|}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{13i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{(23i+188)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{95i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{6\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} \\
& - \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} + \frac{5\,M\gamma^6}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} + \frac{1281\,M\gamma^4 t^{\frac{2}{\gamma}} z^4}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} + \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} + \frac{(16i+5)\;M\gamma^5|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} \\
& + \frac{11i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{(39i-296)\;M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{49i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{6\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{7\,M\gamma^5}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP7 = & -\frac{9\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(3i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{(27i-9)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{15i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(21i+12)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(87i+36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(7i+7)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& - \frac{(9i+21)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(i-35)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(3i-18)\;M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{21\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& - \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{(81i-51)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{53i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(43i+4)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(41i+24)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(15i+15)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(27i+47)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(3i-9)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& - \frac{(i-18)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{3i\,M\gamma^7}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(3i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} \\
& - \frac{3i\,M\gamma^6|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{43\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP8 = & -\frac{9\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(27i-9)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{45i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(39i+36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(9i+21)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{3\,M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{21\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(81i-51)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{185i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& + \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(47i+48)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(27i+47)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(12i-1)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{(6i+11)\;M\gamma^7}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{5\,M\gamma^6|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{43\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& - \frac{(45i-47)\;M\gamma^4 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{223i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{12\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(33i+22)\;M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(26i+26)\;M\gamma^5 t^{\frac{1}{2\gamma}} z}{(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(16i+3)\;M\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{5\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(26i+47)\;M\gamma^6}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{3\,M\gamma^5|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP10 = & \frac{45\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(3i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{(63i-81)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{15i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& + \frac{15i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(129i-36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(i+1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\gamma}} z}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& - \frac{(3i-3)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(7i+19)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(9i+12)\;M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{33\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& - \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{(141i-171)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{131i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& + \frac{7i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(173i-48)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(11i+11)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\gamma}} z}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(i-4)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(5i+13)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(27i+16)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(2i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(6i-3)\;M\gamma^7}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} \\
& - \frac{(3i-2)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(6i-3)\;M\gamma^6|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{25\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{3}{2\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \ldots
\end{aligned}
$$

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\begin{aligned}
NP12 = {} & \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} + \frac{3i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{15i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{5i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} \\
& - \frac{11i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} + \frac{17i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4}{8\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} + \frac{M\gamma^5}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} \\
& - \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3|\gamma-6|}{8\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} + \frac{M\gamma^4|\gamma-6|}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} + \frac{13i\,M\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{10i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{3\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} \\
& + \frac{5\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} - \frac{13\,M\gamma^4}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} + \frac{M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2|\gamma-6|}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} - \frac{M\gamma^3|\gamma-6|}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} \\
& + \frac{5i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{21i\,M\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} + \frac{13i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{6\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^2}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} + \frac{M\gamma^3}{(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} \\
& - \frac{M\gamma^2|\gamma-6|}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2} - \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}t^{\left(\frac{1}{\gamma}\right)} z^2}{(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} + \frac{9i\,M\gamma t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} - \frac{i\,Mt^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{2\,(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^{\frac{7}{2}}} + \frac{3\,M\gamma^2}{(\gamma^3|\gamma-6|-6\,\gamma^2|\gamma-6|)t^2}
\end{aligned}
$$

$$
\begin{aligned}
NP14 = {} & -\frac{45\,M\gamma^7 t^{\frac{2}{\gamma}} z^4}{128\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} - \frac{(63i-81)\;M\gamma^7 t^{\frac{3}{2\gamma}} z^3}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} + \frac{(135i-36)\;M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{15i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{(12i+9)\;M\gamma^7 t^{\frac{1}{2\gamma}} z}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} \\
& + \frac{(33i+21)\;M\gamma^6 t^{\frac{1}{2\gamma}} z|\gamma-6|}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} + \frac{201\,M\gamma^6 t^{\frac{2}{\gamma}} z^4}{64\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} + \frac{(165i-207)\;M\gamma^6 t^{\frac{3}{2\gamma}} z^3}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} - \frac{(771i-204)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} \\
& + \frac{(38i+27)\;M\gamma^6 t^{\frac{1}{2\gamma}} z}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} - \frac{(35i+31)\;M\gamma^5 t^{\frac{1}{2\gamma}} z|\gamma-6|}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} - \frac{M\gamma^7}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{173\,M\gamma^5 t^{\frac{2}{\gamma}} z^4}{32\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} - \frac{(3i+4)\;M\gamma^6|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} \\
& - \frac{(505i-591)\;M\gamma^5 t^{\frac{3}{2\gamma}} z^3}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} + \frac{(703i-188)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} + \frac{7i\,M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \frac{(39i+26)\;M\gamma^5 t^{\frac{1}{2\gamma}} z}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} - \frac{(i-11)\;M\gamma^4 t^{\frac{1}{2\gamma}} z|\gamma-6|}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{11}{4}}} \\
& + \frac{3\,M\gamma^6}{(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} - \frac{143\,M\gamma^4 t^{\frac{2}{\gamma}} z^4}{16\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^5} + \frac{(4i+7)\;M\gamma^5|\gamma-6|}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^2} + \frac{(59i-49)\;M\gamma^4 t^{\frac{3}{2\gamma}} z^3}{2\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{17}{4}}} - \frac{(503i-148)\;M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\,(\gamma^5|\gamma-6|-6\,\gamma^4|\gamma-6|)t^{\frac{7}{2}}} - \dots
\end{aligned}
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\begin{aligned}
NP16 = & -\frac{9\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(3i-3)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{(27i-9)\;M\gamma^6 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{15i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(21i+12)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(87i+36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(7i+7)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(9i+21)\;M\gamma^7 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(i-35)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(3i-18)\;M\gamma^6 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{21\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{(81i-51)\;M\gamma^5 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{53i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(43i+4)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(41i+24)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(15i+15)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\frac{1}{2\,\gamma}} z}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(27i+47)\;M\gamma^6 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(3i-9)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(i-18)\;M\gamma^5 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{3i\,M\gamma^7}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(3i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} \\
& - \frac{3i\,M\gamma^6|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{43\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(i-1)\;M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP17 = & \frac{9\,M\gamma^6 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{128\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} - \frac{(27i-9)\;M\gamma^6 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{45i\,M\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(39i+36)\;M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{64\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{(9i+21)\;M\gamma^7 t^{\frac{1}{2\,\gamma}} z}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{3\,M\gamma^6 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{21\,M\gamma^5 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} + \frac{(81i-51)\;M\gamma^5 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} + \frac{185i\,M\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} \\
& - \frac{i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(47i+48)\;M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{16\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} + \frac{(27i+47)\;M\gamma^6 t^{\frac{1}{2\,\gamma}} z}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} - \frac{(12i-1)\;M\gamma^5 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& - \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^6}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{(6i+11)\;M\gamma^7}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{3\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{5\,M\gamma^6|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{43\,M\gamma^4 t^{\frac{2}{\gamma}} z^4|\gamma-6|}{32\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^5} \\
& - \frac{(45i-47)\;M\gamma^4 t^{\frac{3}{2\,\gamma}} z^3|\gamma-6|}{8\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{17}{4}}} - \frac{223i\,M\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{12\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{3i\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(33i+22)\;M\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{7}{2}}} - \frac{(26i+26)\;M\gamma^5 t^{\frac{1}{2\,\gamma}} z}{(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} \\
& + \frac{(16i+3)\;M\gamma^4 t^{\frac{1}{2\,\gamma}} z|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^{\frac{11}{4}}} + \frac{5\,M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^5}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{(26i+47)\;M\gamma^6}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \frac{M\sqrt{3\,\gamma-2}\sqrt{\gamma-2}\gamma^4|\gamma-6|}{4\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} - \frac{3\,M\gamma^5|\gamma-6|}{2\,(\gamma^6-12\,\gamma^5+36\,\gamma^4)t^2} + \ldots
\end{aligned}
$$

Bianchi Identities

$$
\begin{aligned}
BI2 = & -\frac{135\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{2}{\gamma}}z^{4}\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}+\frac{9\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{2}{\gamma}}z^{4}|\gamma-6|^{\frac{3}{2}}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}-\frac{(315i-315)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\frac{3}{2\gamma}}z^{3}\sqrt{|\gamma-6|}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}} \\
& +\frac{(405i+135)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{3}{2\gamma}}z^{3}\sqrt{|\gamma-6|}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}}-\frac{(3i-3)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\frac{3}{2\gamma}}z^{3}|\gamma-6|^{\frac{3}{2}}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}}-\frac{(9i-9)\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{3}{2\gamma}}z^{3}|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}} \\
& -\frac{(189i+270)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\left(\frac{1}{\gamma}\right)}z^{2}\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}+\frac{(105i+81)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\left(\frac{1}{\gamma}\right)}z^{2}\sqrt{|\gamma-6|}}{32\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}+\frac{(9i+6)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\left(\frac{1}{\gamma}\right)}z^{2}|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}} \\
& +\frac{327i\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\left(\frac{1}{\gamma}\right)}z^{2}|\gamma-6|^{\frac{3}{2}}}{64\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}+\frac{(29i-403)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\frac{1}{2\gamma}}z\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}-\frac{(39i-657)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{1}{2\gamma}}z\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}} \\
& -\frac{(337i+337)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\frac{1}{2\gamma}}z|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}+\frac{(123i+291)\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{1}{2\gamma}}z|\gamma-6|^{\frac{3}{2}}}{64\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}+\frac{513\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{2}{\gamma}}z^{4}\sqrt{|\gamma-6|}}{32\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}-\ldots
\end{aligned}
$$

$$
\begin{aligned}
BI3 = & \frac{135\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{2}{\gamma}}z^{4}\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}-\frac{9\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{2}{\gamma}}z^{4}|\gamma-6|^{\frac{3}{2}}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}-\frac{(315i-315)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\frac{3}{2\gamma}}z^{3}\sqrt{|\gamma-6|}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}} \\
& +\frac{(405i+135)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{3}{2\gamma}}z^{3}\sqrt{|\gamma-6|}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}}-\frac{(3i-3)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\frac{3}{2\gamma}}z^{3}|\gamma-6|^{\frac{3}{2}}}{256\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}}-\frac{(9i-9)\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{3}{2\gamma}}z^{3}|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{21}{4}}} \\
& +\frac{(189i+270)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\left(\frac{1}{\gamma}\right)}z^{2}\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}-\frac{(105i+81)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\left(\frac{1}{\gamma}\right)}z^{2}\sqrt{|\gamma-6|}}{32\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}-\frac{(9i+6)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\left(\frac{1}{\gamma}\right)}z^{2}|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}} \\
& -\frac{327i\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\left(\frac{1}{\gamma}\right)}z^{2}|\gamma-6|^{\frac{3}{2}}}{64\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{9}{2}}}+\frac{(29i-403)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{7}t^{\frac{1}{2\gamma}}z\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}-\frac{(39i-657)\sqrt{2}M^{\frac{3}{2}}\gamma^{8}t^{\frac{1}{2\gamma}}z\sqrt{|\gamma-6|}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}} \\
& -\frac{(337i+337)\sqrt{2}M^{\frac{3}{2}}\sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^{6}t^{\frac{1}{2\gamma}}z|\gamma-6|^{\frac{3}{2}}}{128\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}+\frac{(123i+291)\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{1}{2\gamma}}z|\gamma-6|^{\frac{3}{2}}}{64\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{\frac{15}{4}}}-\frac{513\sqrt{2}M^{\frac{3}{2}}\gamma^{7}t^{\frac{2}{\gamma}}z^{4}\sqrt{|\gamma-6|}}{32\left(\gamma^{6}|\gamma-6|-12\gamma^{5}|\gamma-6|+36\gamma^{4}|\gamma-6|\right)t^{6}}+\ldots
\end{aligned}
$$

$$
\begin{aligned}
BI5 = {} & \frac{135\, M^{\frac{3}{2}} \gamma^8 t^{\frac{2}{\gamma}} z^4}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6 \sqrt{|\gamma-6|}} + \frac{9\, M^{\frac{3}{2}} \gamma^7 t^{\frac{2}{\gamma}} z^4 \sqrt{|\gamma-6|}}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6} - \frac{(81i-81)\ M^{\frac{3}{2}} \gamma^8 t^{\frac{3}{2\gamma}} z^3}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}} \sqrt{|\gamma-6|}} \\
& - \frac{(9i-9)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{3}{2\gamma}} z^3 \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}}} - \frac{99i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} + \frac{81i\, M^{\frac{3}{2}} \gamma^8 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} \\
& + \frac{3i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} + \frac{93i\, M^{\frac{3}{2}} \gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} - \frac{(39i+39)\ M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^7 t^{\frac{1}{2\gamma}} z}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}} \sqrt{|\gamma-6|}} \\
& + \frac{(63i+63)\ M^{\frac{3}{2}} \gamma^8 t^{\frac{1}{2\gamma}} z}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}} \sqrt{|\gamma-6|}} + \frac{(3i+3)\ M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\gamma}} z \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}}} \\
& + \frac{(87i+3)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{1}{2\gamma}} z \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}}} - \frac{189\, M^{\frac{3}{2}} \gamma^7 t^{\frac{2}{\gamma}} z^4}{16\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6 \sqrt{|\gamma-6|}} - \frac{21\, M^{\frac{3}{2}} \gamma^6 t^{\frac{2}{\gamma}} z^4 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6} \\
& + \frac{(405i-405)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{3}{2\gamma}} z^3}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}} \sqrt{|\gamma-6|}} + \frac{(51i-51)\ M^{\frac{3}{2}} \gamma^6 t^{\frac{3}{2\gamma}} z^3 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}}} + \frac{435i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} \\
& - \frac{99i\, M^{\frac{3}{2}} \gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} - \frac{13i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} - \frac{183i\, M^{\frac{3}{2}} \gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{16\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI6 = {} & \frac{135\, M^{\frac{3}{2}} \gamma^8 t^{\frac{2}{\gamma}} z^4}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6 \sqrt{|\gamma-6|}} + \frac{9\, M^{\frac{3}{2}} \gamma^7 t^{\frac{2}{\gamma}} z^4 \sqrt{|\gamma-6|}}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6} + \frac{(81i-81)\ M^{\frac{3}{2}} \gamma^8 t^{\frac{3}{2\gamma}} z^3}{128\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}} \sqrt{|\gamma-6|}} \\
& + \frac{(9i-9)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{3}{2\gamma}} z^3 \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}}} - \frac{99i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} + \frac{81i\, M^{\frac{3}{2}} \gamma^8 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} \\
& + \frac{3i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} + \frac{93i\, M^{\frac{3}{2}} \gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} + \frac{(39i+39)\ M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^7 t^{\frac{1}{2\gamma}} z}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}} \sqrt{|\gamma-6|}} \\
& - \frac{(63i+63)\ M^{\frac{3}{2}} \gamma^8 t^{\frac{1}{2\gamma}} z}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}} \sqrt{|\gamma-6|}} - \frac{(3i+3)\ M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\frac{1}{2\gamma}} z \sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}}} \\
& - \frac{(87i+3)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{1}{2\gamma}} z \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{15}{4}}} - \frac{189\, M^{\frac{3}{2}} \gamma^7 t^{\frac{2}{\gamma}} z^4}{16\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6 \sqrt{|\gamma-6|}} - \frac{21\, M^{\frac{3}{2}} \gamma^6 t^{\frac{2}{\gamma}} z^4 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^6} \\
& - \frac{(405i-405)\ M^{\frac{3}{2}} \gamma^7 t^{\frac{3}{2\gamma}} z^3}{64\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}} \sqrt{|\gamma-6|}} - \frac{(51i-51)\ M^{\frac{3}{2}} \gamma^6 t^{\frac{3}{2\gamma}} z^3 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{21}{4}}} + \frac{435i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} \\
& - \frac{99i\, M^{\frac{3}{2}} \gamma^7 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}} \sqrt{|\gamma-6|}} - \frac{13i\, M^{\frac{3}{2}} \sqrt{3\gamma-2}\sqrt{\gamma-2}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} - \frac{183i\, M^{\frac{3}{2}} \gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2 \sqrt{|\gamma-6|}}{16\left(\sqrt{2}\gamma^6 - 12\sqrt{2}\gamma^5 + 36\sqrt{2}\gamma^4\right) t^{\frac{9}{2}}} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI9 = {} & \frac{9\,M^{\frac{3}{2}}\gamma^5 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{64\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^6} - \frac{(9i-9)\;M^{\frac{3}{2}}\gamma^5 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{21}{4}}} + \frac{21i\,M^{\frac{3}{2}}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{64\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} - \frac{3i\,M^{\frac{3}{2}}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{16\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}} + \frac{3i\,M^{\frac{3}{2}}\gamma^6 t^{\frac{1}{2\gamma}} z}{16\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} \\
& + \frac{(15i+3)\;M^{\frac{3}{2}}\gamma^5 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{16\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}} - \frac{15\,M^{\frac{3}{2}}\gamma^4 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{32\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^6} + \frac{(3i-3)\;M^{\frac{3}{2}}\gamma^4 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{21}{4}}} - \frac{47i\,M^{\frac{3}{2}}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{16\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} + \frac{19i\,M^{\frac{3}{2}}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{8\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}} \\
& - \frac{13i\,M^{\frac{3}{2}}\gamma^5 t^{\frac{1}{2\gamma}} z}{8\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} - \frac{(5i+1)\;M^{\frac{3}{2}}\gamma^4 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}} - \frac{3\,M^{\frac{3}{2}}\gamma^6}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^3\sqrt{|\gamma-6|}} - \frac{3\,M^{\frac{3}{2}}\gamma^5\sqrt{|\gamma-6|}}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^3} - \frac{M^{\frac{3}{2}}\gamma^3 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{8\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^6} \\
& - \frac{(11i-11)\;M^{\frac{3}{2}}\gamma^3 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{21}{4}}} + \frac{13i\,M^{\frac{3}{2}}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} - \frac{21i\,M^{\frac{3}{2}}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}} + \frac{13i\,M^{\frac{3}{2}}\gamma^4 t^{\frac{1}{2\gamma}} z}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} + \frac{(5i+1)\;M^{\frac{3}{2}}\gamma^3 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{15}{4}}} \\
& + \frac{9\,M^{\frac{3}{2}}\gamma^5}{\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^3\sqrt{|\gamma-6|}} + \frac{7\,M^{\frac{3}{2}}\gamma^2 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^6} + \frac{(2i-2)\;M^{\frac{3}{2}}\gamma^2 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{21}{4}}} - \frac{17i\,M^{\frac{3}{2}}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} + \frac{5i\,M^{\frac{3}{2}}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{2\left(\sqrt{2}\gamma^4-6\sqrt{2}\gamma^3\right)t^{\frac{9}{2}}} - \dots
\end{aligned}
$$

$$
\begin{aligned}
BI11 = {} & \frac{9\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{128\left(\gamma^4-6\gamma^3\right)t^6} + \frac{(9i-9)\;\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{64\left(\gamma^4-6\gamma^3\right)t^{\frac{21}{4}}} + \frac{21i\sqrt{2}M^{\frac{3}{2}}\gamma^6 t^{\left(\frac{1}{\gamma}\right)} z^2}{128\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} - \frac{3i\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{32\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}} - \frac{3i\sqrt{2}M^{\frac{3}{2}}\gamma^6 t^{\frac{1}{2\gamma}} z}{32\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} \\
& - \frac{(15i+3)\;\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{32\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}} - \frac{15\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{64\left(\gamma^4-6\gamma^3\right)t^6} - \frac{(3i-3)\;\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{21}{4}}} - \frac{47i\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\left(\frac{1}{\gamma}\right)} z^2}{32\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} + \frac{19i\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{16\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}} \\
& + \frac{13i\sqrt{2}M^{\frac{3}{2}}\gamma^5 t^{\frac{1}{2\gamma}} z}{16\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} + \frac{(5i+1)\;\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}} - \frac{3\sqrt{2}M^{\frac{3}{2}}\gamma^6}{4\left(\gamma^4-6\gamma^3\right)t^3\sqrt{|\gamma-6|}} - \frac{3\sqrt{2}M^{\frac{3}{2}}\gamma^5\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^3} - \frac{\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{16\left(\gamma^4-6\gamma^3\right)t^6} \\
& + \frac{(11i-11)\;\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{21}{4}}} + \frac{13i\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\left(\frac{1}{\gamma}\right)} z^2}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} - \frac{21i\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}} - \frac{13i\sqrt{2}M^{\frac{3}{2}}\gamma^4 t^{\frac{1}{2\gamma}} z}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} - \frac{(5i+1)\;\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\frac{1}{2\gamma}} z\sqrt{|\gamma-6|}}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}} \\
& + \frac{9\sqrt{2}M^{\frac{3}{2}}\gamma^5}{2\left(\gamma^4-6\gamma^3\right)t^3\sqrt{|\gamma-6|}} + \frac{7\sqrt{2}M^{\frac{3}{2}}\gamma^2 t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{8\left(\gamma^4-6\gamma^3\right)t^6} - \frac{(i-1)\;\sqrt{2}M^{\frac{3}{2}}\gamma^2 t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{\left(\gamma^4-6\gamma^3\right)t^{\frac{21}{4}}} - \frac{17i\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} + \frac{5i\sqrt{2}M^{\frac{3}{2}}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}} \\
& + \frac{3i\sqrt{2}M^{\frac{3}{2}}\gamma^3 t^{\frac{1}{2\gamma}} z}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{15}{4}}\sqrt{|\gamma-6|}} - \frac{7\sqrt{2}M^{\frac{3}{2}}\gamma t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{8\left(\gamma^4-6\gamma^3\right)t^6} + \frac{(i-1)\;\sqrt{2}M^{\frac{3}{2}}\gamma t^{\frac{3}{2\gamma}} z^3\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^{\frac{21}{4}}} + \frac{3i\sqrt{2}M^{\frac{3}{2}}\gamma^2 t^{\left(\frac{1}{\gamma}\right)} z^2}{8\left(\gamma^4-6\gamma^3\right)t^{\frac{9}{2}}\sqrt{|\gamma-6|}} + \frac{\sqrt{2}M^{\frac{3}{2}} t^{\frac{2}{\gamma}} z^4\sqrt{|\gamma-6|}}{4\left(\gamma^4-6\gamma^3\right)t^6}
\end{aligned}
$$

Petrov Type

Type "I"